DAY 43

Detecting Concept Drift & Model Performance Degradation

60-Day Data Science Journey | Practical Python Guide

Objective: Learn drift types, calculate MAE over time, and interpret a monitoring warning without treating it as proof of a cause.

Tools: Python, pandas, NumPy

1. The main concepts

Data drift: the input feature distribution changes. Example: the average age of incoming customers changes.

Concept drift: the relationship between features and target changes. Example: discounts no longer affect purchases in the same way.

Performance degradation: measured model performance becomes worse. Example: regression MAE increases on evaluated data.

Roman Urdu: Data drift mein input data ki distribution badalti hai. Concept drift mein input aur target ka relation badalta hai. Performance degradation ka matlab evaluation par errors barhna hai.

2. Mean Absolute Error (MAE)

MAE is the average absolute difference between actual values and predictions. It is a regression metric and uses the target's units.

```
MAE = average(|actual − predicted|)
```

Example: actual = [10, 20, 30], predictions = [12, 18, 33]. Absolute errors are [2, 2, 3], so MAE = 7 / 3 ≈ 2.33.

Labels matter: MAE needs actual outcomes. If labels are delayed, you cannot yet calculate a supervised performance metric for those predictions.

3. Hands-on Python example

```
import pandas as pd

data = pd.DataFrame({
    "period": ["Week 1"] * 5 + ["Week 2"] * 5,
    "actual": [100, 120, 130, 110, 140,
               100, 120, 130, 110, 140],
    "predicted": [102, 118, 128, 112, 138,
                  115, 145, 100, 140, 165],
})
data["absolute_error"] = (
    data["actual"] - data["predicted"]
).abs()
print(data.groupby("period")["absolute_error"].mean())
```

Expected output

```
Week 1     2.0
Week 2    25.0
```

Week 2 MAE is 23 points higher than Week 1. With an illustrative threshold of 10, the monitor raises a warning. The threshold is an example, not a universal rule.

4. What to investigate

• Confirm actual labels and predictions are correctly aligned.

• Check missing values, unexpected categories, units, ranges, and data-pipeline changes.

• Consider seasonality, population changes, and known business events.

• Evaluate on representative samples and investigate before retraining.

Do not overclaim: higher MAE indicates larger measured errors on this sample. It does not by itself prove concept drift, identify the cause, or guarantee retraining will help.

5. Step-by-step code explanation

Create the data: each row has a period, actual value, and prediction.

Absolute errors: subtract prediction from actual and use abs() to make the error magnitude non-negative.

Group by period: groupby() groups rows from the same week, and mean() calculates MAE.

Compare baseline: subtract baseline MAE from latest MAE. If the increase exceeds a chosen threshold, flag it for investigation.

6. Exercises with solutions

Exercise 1: Actual [8, 10, 12], predicted [7, 12, 11]. **Solution:** Absolute errors [1, 2, 1], so MAE = 4/3 ≈ 1.33.

Exercise 2: Baseline MAE 4.5, latest MAE 9.0. **Solution:** Increase = 4.5. Whether to alert depends on the monitoring policy.

Exercise 3: Inputs changed but MAE stayed similar. **Solution:** Drift in inputs does not automatically prove performance degradation.

Exercise 4: MAE increased. Should you retrain immediately? **Solution:** No. Validate data and labels, investigate likely causes, then choose an appropriate response.

7. Run the practice file

```
pip install pandas numpy
python Day-43-Concept-Drift-Monitoring-Practice.py
python Day-43-Concept-Drift-Monitoring-Practice.py --test
```

The practice file includes validation for required columns and missing values, plus built-in assertions for expected MAE values and warning behavior.

8. Knowledge check

1. Which drift means the input-target relationship changes?
2. What does MAE rising from 2 to 25 indicate?
3. Does data drift always mean performance declined?
4. Why do delayed labels complicate monitoring?

Answer key: 1. Concept drift. 2. Average absolute error increased on evaluated data. 3. No, evaluate performance separately. 4. Without actual outcomes, supervised metrics cannot yet be calculated.

9. Summary

You learned the differences between data drift, concept drift, and performance degradation, and practiced monitoring MAE over time. Treat alerts as signals to investigate, not automatic diagnoses.

Repository: https://github.com/DeveloperDesigner-star/Datasciencepdfs